# Prolog-RAG: A Symbolic Reasoning Approach to Retrieval-Augmented Generation

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***Abstract*—Retrieval-Augmented Generation (RAG) enhances large language models (LLMs) by incorporating external knowledge, but its application in specialized domains like environmental protection remains limited due to challenges in logical reasoning and knowledge updates. We propose Prolog-RAG, a lightweight framework that integrates Prolog-based symbolic reasoning with LLMs to support domain-specific tasks. Prolog enables structured, rule-based inference with high interpretability and low maintenance cost. We develop a Prolog-RAG system for real-world environmental scenarios, such as wastewater diagnostics and policy queries, where precise logic and timely knowledge are essential. Experimental results on benchmark datasets demonstrate that Prolog-RAG significantly improves answer relevance over traditional vector-based RAG methods, particularly in logic-driven and frequently changing domains. This work offers a practical approach to combining symbolic reasoning with generative models, providing a new direction for retrieval-augmented generation in specialized applications.**



***Index Terms*—RAG, Prolog, Logical Reasoning, Environmental AI, Wastewater Treatment, LLMs**


## I. Introduction

As global challenges such as water pollution and resource scarcity intensify, intelligent and data- driven approaches have become essential to sustainable water management. Smart water systems that integrate IoT, cloud computing, and AI are increasingly adopted to support precise and efficient wastewater treatment. Aligned with the United Nations' Sustainable Development Goals (SDGs), many governments are promoting digital modernization of water governance [1].

In China, significant progress has been made in building intelligent wastewater infrastructure. Online monitoring, anomaly detection, and AI-driven decision systems are now widely deployed, enabling the transition from manual, experience-based operations to data-centric management [2]. In major cities such as Beijing and Shanghai, machine learning models are actively used for water quality prediction, aeration control, and equipment diagnostics. However, existing systems still face limitations in integrating domain knowledge, adapting to evolving regulations, and supporting transparent reasoning—challenges that are critical in complex and safety-sensitive scenarios [3]–[5].

Retrieval-Augmented Generation (RAG) offers a promising solution by combining external knowledge retrieval with the generative power of LLMs [6], [7]. While successful in open-domain applications, conventional RAG struggles in specialized domains like environmental protection, where domain expertise, logic consistency, and real-time knowledge updates are essential. Recent efforts have introduced structured knowledge sources such as knowledge graphs into RAG pipelines, enabling more factual and interpretable generation [8]–[10]. Yet these approaches often require costly infrastructure and expert curation, limiting their adoption in small and mediumsized enterprises (SMEs). In contrast, symbolic systems like Prolog provide a lightweight, rule- based alternative that offers high interpretability and low maintenance.

In this work, we propose Prolog-RAG, a domain-specific RAG framework that integrates Prolog reasoning [11]–[13] with LLMs. Designed for scenarios such as wastewater diagnostics, staff training, and policy interpretation, the system bridges symbolic and neural approaches to deliver accurate, explainable, and updatable responses. Experiments on benchmark datasets demonstrate its advantages in relevance and logic alignment compared to standard RAG systems.


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## II. Related Works

### A. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) enhances large language models (LLMs) by retrieving external knowledge to improve factual accuracy and adaptability. Foundational models such as REALM [14] and RAG [6] introduced joint architectures where neural retrievers supply contextual passages to a generative model. These systems significantly improved knowledge-intensive NLP tasks by enabling dynamic access to information during inference. Subsequent research improved retrieval quality through dense representations and late interaction mechanisms, such as ColBERT [15]. However, these approaches rely heavily on unstructured text retrieval and lack support for structured reasoning or logical constraints—making them less suitable for specialized tasks that require interpretable, rule-based logic.

### B. Structured Knowledge in RAG

To address the limitations of unstructured retrieval, researchers have explored integrating structured knowledge, particularly knowledge graphs, into RAG pipelines. Some approaches retrieve subgraphs and perform multi-hop reasoning to improve factual consistency, while others employ graph neural networks (GNNs) [16] to propagate and embed structural knowledge. Graph-RAG [10] and similar models [17] combine knowledge graphs with LLMs through graph traversal or semantic expansion. Although these methods improve reasoning over structured data, they typically involve high construction costs, manual curation, and computational overhead, limiting their scalability in domain-specific or SME contexts. More adaptive methods like Path-RAG [18] attempt to reduce complexity, but the overall barrier to practical deployment remains high.

### C. Logic Programming and Neuro-Symbolic Reasoning

Parallel to graph-based methods, logic programming offers a rule-driven alternative for integrating symbolic reasoning into language generation. Prolog, a declarative logic programming language, supports transparent inference via unification and backtracking, and has long been used in expert systems and knowledge engineering. Recent efforts have combined Prolog-style reasoning with neural models. Differentiable logic frameworks enable neural networks to learn over logical rules [12], [13], improving interpretability and compositionality. While promising, these neuro-symbolic methods have not yet been fully integrated into RAG architectures. Most existing RAG systems still lack traceable reasoning paths, which are essential for tasks requiring explicit logic, such as policy compliance or fault diagnosis in industrial domains.

## III. Methodology

### A. Overview

We propose Prolog-RAG, a hybrid framework that integrates symbolic reasoning with generative models for domain-specific applications. The system addresses key challenges in existing RAG systems by combining logic-driven inference with LLM-based language generation. The workflow includes: (1) query parsing, (2) symbolic inference via Prolog, (3) response generation using an LLM, and (4) semantic scoring for candidate selection.

### B. System Architecture

As shown in Fig. 1, the system consists of four core components: (1) User Interface: Accepts natural language input, performs entity extraction and intent classification, and transforms the query into a Prolog-compatible form; (2) Prolog Knowledge Base: Stores structured domain knowledge as facts and rules. The Prolog engine executes logical inference based on user queries, producing explicit reasoning results. (3) LLM Generator: Receives Prolog outputs embedded into prompt templates and generates natural language responses. (4) Scoring Module: Ranks generated candidates based on semantic alignment with both the original query and Prolog inference output. This modular architecture enables flexible and interpretable domain reasoning without sacrificing generation fluency.

### C. Knowledge Construction and Logical Inference

As shown in Fig. 2, the Prolog knowledge base is derived from domain texts through semantic parsing and rule extraction. A typical rule is expressed as:

```
increase_blower_speed(Tank):
- oxygen_level(Tank, O), O < 2.
```

User queries are mapped to Prolog syntax and evaluated via unification and backward chaining. For example:

```
?- anomaly(aeration_tank).
```

yields a reasoning result if the preconditions in the rule base are met. This ensures consistent and traceable logic paths throughout the system.

### D. Prompted Generation and Response Ranking

Each Prolog result is embedded into a prompt template and passed to the LLM to generate multiple responses. We rank these candidates using a composite similarity score:

$$S_i = \alpha \cdot \text{sim}(q, rli) + \beta \cdot \text{sim}(r_p, rli) \tag{1}$$

where $q$ is the original query, $r_p$ is the Prolog output, and $r_{li}$ is a generated candidate. The weights $\alpha$ and $\beta$ are adjusted based on the task—placing higher emphasis on logical fidelity for diagnostics and on relevance for general queries. The top-ranked response is returned to the user as the final output, ensuring both semantic alignment and logical consistency.

## IV. Experimental Evaluation

### A. Experimental Setup

We compare Prolog-RAG against a baseline Naive RAG system that uses dense vector retrieval with the same LLM backend (GLM-4-air). Both systems share decoding parameters and operate under identical hardware configurations

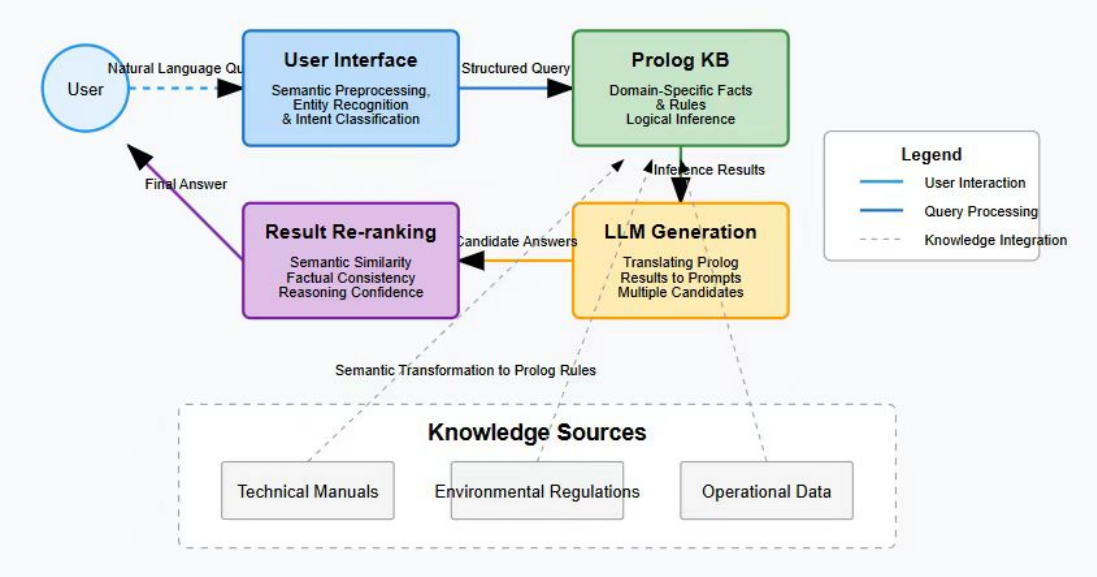



Fig. 1. Prolog-RAG system architecture.

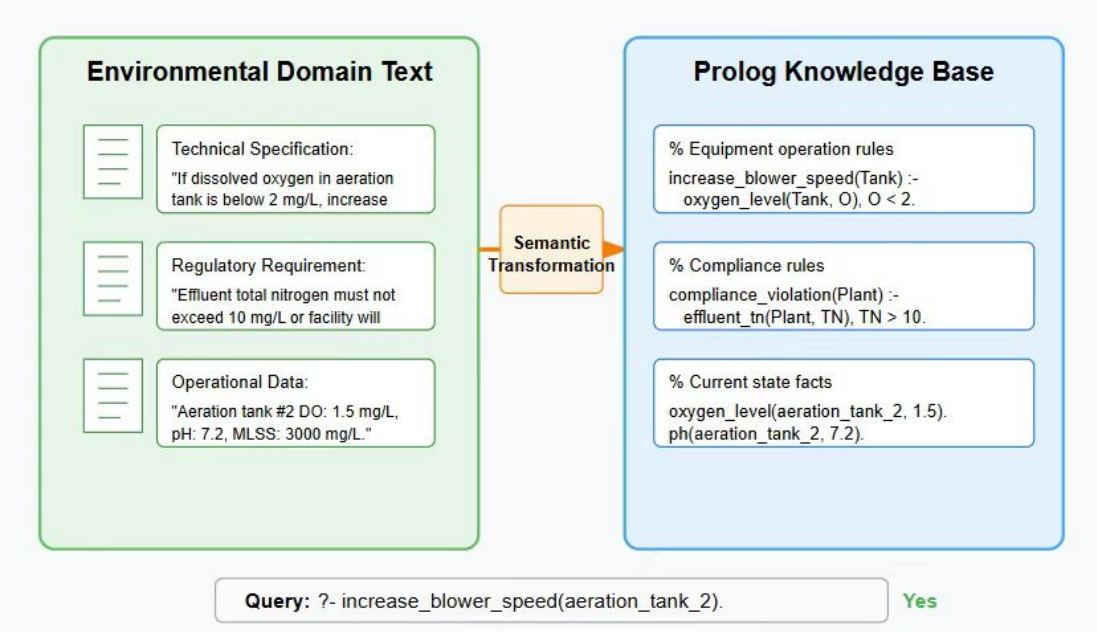



Fig. 2. Prolog knowledge representation and reasoning.

(NVIDIA A100 GPU, 256GB RAM). Prolog-RAG performs logic-driven inference via a structured Prolog knowledge base, while Naive RAG retrieves top-k unstructured text segments from a vector store to prompt the LLM.

### *B. Datasets*

We use three representative datasets from RAGBench:

- **CovidQA**: 246 biomedical questions of COVID-19.
- **DelucionQA**: 184 technical support questions derived from industrial troubleshooting manuals.
- **HotpotQA (subset)**: 390 open-domain questions requiring multi-hop reasoning.

For fairness, all datasets are converted into both structured Prolog rules and segmented text for retrieval-based comparison.

### *C. Evaluation Metrics*

We evaluate using four metrics:

- **Answer Relevance (AR)**: Semantic similarity between the user query and generated response.
- **Faithfulness (FA)**: Degree to which the answer remains grounded in source knowledge.
- **Completeness (CO)**: Coverage of necessary facts for a full answer.

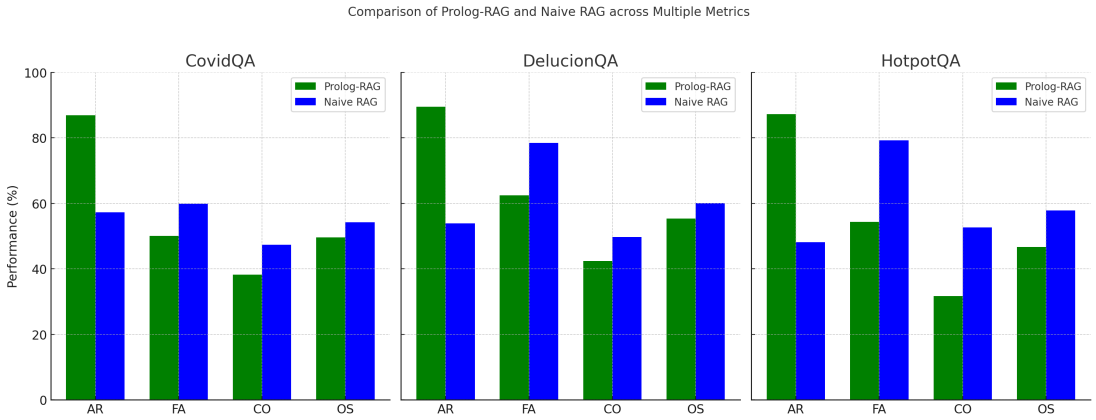



Fig. 3. Comparison of Prolog-RAG and Navie RAG on some benchmark datasets.

- **Overall Score (OS)**: Weighted harmonic mean of AR, FA, and CO.

### D. Results and Analysis

Table 1 summarizes the results of both systems across the three datasets:

TABLE I
COMPARISON OF PROLOG-RAG AND NAIVE RAG ACROSS THREE DATASETS

| Dataset | Method | AR (%) | FA (%) | CO (%) | OS (%) |
|---|---|---|---|---|---|
| CovidQA | Prolog-RAG | **86.9** | 50.1 | 38.2 | 49.6 |
| CovidQA | Naive RAG | 57.3 | **59.8** | **47.3** | **54.2** |
| DelucionQA | Prolog-RAG | **89.5** | 62.5 | 42.4 | 55.3 |
| DelucionQA | Naive RAG | 53.9 | **78.4** | **49.7** | **60.1** |
| HotpotQA | Prolog-RAG | **87.2** | 54.3 | 31.7 | 46.7 |
| HotpotQA | Naive RAG | 48.1 | **79.2** | **52.7** | **57.8** |

The comparison can be further visualized in Fig. 3. Based on the experimental results, we observe that Prolog-RAG consistently outperforms Naive RAG in Answer Relevance (AR) across all evaluated datasets, with margins exceeding 30 percentage points in some cases. This indicates that Prolog-RAG is more effective at capturing user intent through symbolic reasoning and structured knowledge integration. In contrast, Naive RAG achieves higher scores in Faithfulness (FA) and Completeness (CO), primarily due to its reliance on directly extractable content from source documents. Although Prolog-RAG's logic-based transformation introduces a level of abstraction that may lead to minor information loss, it remains competitive in terms of Overall Score (OS). Notably, Prolog-RAG surpasses Naive RAG on DelucionQA—the dataset most closely aligned with our environmental use cases—highlighting its strong potential for real-world industrial applications. These results validate that Prolog-RAG offers a better trade-off between precision, reasoning traceability, and relevance, even if minor losses in low-level detail occur due to rule-based abstraction.

## V. DISCUSSION

The experimental results demonstrate that Prolog-RAG is a promising framework for domain-specific reasoning tasks that require a balance between factual relevance, logical consistency, and interpretability. Compared to traditional vector-based RAG systems, Prolog-RAG offers several notable advantages, particularly in scenarios where logical rigor and up-to-date knowledge are critical.

### A. Strengths and Practical Value

One of the key strengths of Prolog-RAG lies in its symbolic reasoning capability. The Prolog knowledge base encodes domain knowledge using explicit rules and facts, enabling the system to perform transparent, traceable inference rather than relying solely on statistical associations. This allows Prolog-RAG to better understand user intent and generate responses that are logically consistent, which is reflected in its superior performance in answer relevance (AR) across all tested datasets. Another practical advantage is the lightweight and maintainable nature of Prolog-based knowledge bases. Unlike knowledge graphs, which often require extensive manual curation and complex ontological design, Prolog rules can be incrementally added and updated with minimal effort. This makes the system highly suitable for small and medium-sized enterprises (SMEs) in domains like environmental services, where technical resources are limited but precise logic is essential. Furthermore, the integration with LLMs enhances the naturalness and fluency of the final responses. By using Prolog results as input to an LLM prompt, the system combines symbolic reasoning with generative flexibility, producing answers that are both accurate and user-friendly.

### B. Limitations and Areas for Improvemente

Despite its strengths, Prolog-RAG exhibits certain limitations, particularly in faithfulness and completeness. These shortcomings arise primarily from the abstraction and compression that occur when transforming unstructured domain texts into formal logic rules. While Prolog rules capture the essence of the knowledge, some fine-grained details or contextual nuances may be lost in the process. This can result in responses that, although logically valid, may omit specific factual elements found in the original source. To address

this, future work should explore more robust rule extraction methods, such as semantic role labeling combined with neural symbolic parsing, to preserve more contextual information. Additionally, hybrid systems that combine symbolic reasoning with vector-based retrieval could be developed to achieve complementary strengths—retrieving supporting passages for completeness while using logic rules to ensure reasoning validity.

### C. Applicability in Environmental and Specialized Domains

The proposed system is especially well-suited for use in environmental operations, such as wastewater treatment, where logic-based rules govern equipment behavior, anomaly diagnosis, and compliance with evolving regulatory standards. Moreover, the system's architecture is domain-agnostic, meaning it can be adapted to other professional fields such as industrial safety, energy management, and even medical diagnostics—where knowledge is structured, decisions must be justified, and knowledge updates are frequent. By bridging symbolic AI and neural language generation, Prolog-RAG offers a deployable and extensible solution for enterprises seeking trustworthy AI tools in logic-intensive environments. Its explainable nature also makes it valuable for training, auditing, and decision support, areas where black-box models often fall short.

## VI. Conclusion

This paper introduces Prolog-RAG, a logic-based retrieval-augmented generation framework that integrates Prolog symbolic reasoning with large language models (LLMs) for specialized domains. Unlike traditional vector-based RAG systems, Prolog-RAG leverages structured rules and interpretable inference to improve response relevance, especially in scenarios requiring precise reasoning and dynamic knowledge updates. Through experiments on benchmark datasets, we show that Prolog-RAG outperforms conventional RAG approaches in terms of semantic alignment and logical consistency, while maintaining flexibility in language generation. Its low-cost maintenance and ease of domain adaptation make it particularly suitable for applications in environmental management and similar fields. Although some limitations remain in terms of factual completeness—mainly due to rule abstraction—these can be addressed by enhancing rule extraction and incorporating hybrid retrieval strategies. Overall, this work provides a lightweight, interpretable, and domain-adaptable RAG framework, paving the way for integrating symbolic logic with neural generation in real-world industrial systems.

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